Advanced Survival Analysis
Veteran Affairs Lung Cancer Dataset

Group 1

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Introduction

What we found in Assignment 1

- We modeled survival using parametric models such as Gompertz, Exponential, etc..
- The generalized gamma model gave the best fit among the parametric models.
- Survival time was highly variable, with many early deaths and a small number of long survivors.
- The hazard was not constant over time and declined after the early period.
- The first presentation mainly described the overall survival pattern and compared parametric models.

This presentation presents the use of non-parametric and semi-parametric survival methods.

Objective and Research Questions

Objective

To evaluate survival time and the risk of death in lung cancer patients in Veterans, and identify key factors.

Research Questions

1. How does the Kaplan–Meier survival curve compare with the survival curve from the best-fitting parametric model?
2. How does the Nelson–Aalen hazard trend compare with the hazard function implied by the best-fitting parametric model?
3. Do the Log-rank and Wilcoxon test results agree with the univariable Cox model results?
4. What is the overall survival experience of the study population, including the median survival time and key survival probabilities?
5. What is the nature of the hazard over time: increasing, decreasing, constant, or non-monotonic?
6. Which covariates are the key predictors of survival in the final Cox model, and how should their hazard ratios be interpreted?
7. What are the clinical or practical implications of these findings?

Methodology

Data

Source: Veteran Affairs Lung Cancer Study

Sample: 137 patients with- 128 deaths and 9 censored observations Main

variables: survival time, event status, treatment, age, cell type, prior therapy, Karnofsky score

Methods used

- Kaplan–Meier survival curves and percentiles
- Comparison with the generalized gamma model
- Nelson–Aalen cumulative hazard and hazard trend
- Log-rank and Breslow tests
- Univariable and multivariable Cox models
- PH and Martingale diagnostics
- Refined Cox model with cell type stratification and time-varying Karnofsky score

Kaplan–Meier Survival Estimates

Table 1: Standard therapy survival estimate

time	n.risk	n.event	survival	CI
3	69	1	0.986	0.958 - 1
4	68	1	0.971	0.932 - 1
7	67	1	0.957	0.91 - 1
8	66	2	0.928	0.868 - 0.991
10	64	2	0.899	0.83 - 0.973
11	62	1	0.884	0.812 - 0.963
12	61	2	0.855	0.776 - 0.942
13	59	1	0.841	0.758 - 0.932
16	58	1	0.826	0.741 - 0.921
18	57	2	0.797	0.708 - 0.898
20	55	1	0.783	0.691 - 0.886
21	54	1	0.768	0.675 - 0.874
22	53	1	0.754	0.659 - 0.862
27	51	1	0.739	0.642 - 0.85
30	50	1	0.724	0.626 - 0.838

Table 2: Test therapy survival estimate

time	n.risk	n.event	survival	CI
1	68	2	0.971	0.931 - 1
2	66	1	0.956	0.908 - 1
7	65	2	0.926	0.866 - 0.991
8	63	2	0.897	0.828 - 0.972
13	61	1	0.882	0.809 - 0.962
15	60	2	0.853	0.773 - 0.941
18	58	1	0.838	0.755 - 0.93
19	57	2	0.809	0.721 - 0.908
20	55	1	0.794	0.704 - 0.896
21	54	1	0.779	0.687 - 0.884
24	53	2	0.750	0.654 - 0.86
25	51	3	0.706	0.605 - 0.823
29	48	1	0.691	0.59 - 0.81
30	47	1	0.676	0.574 - 0.797
31	46	1	0.662	0.558 - 0.784

Non-Parametric Analysis – Kaplan–Meier Estimator

Table 3: Survival percentiles

Group	25%	50%	75%
Overall	25 (19 - 36)	80 (52 - 105)	162 (133 - 242)
Standard	27 (16 - 54)	103 (59 - 132)	162 (139 - 260)
Test	24.5 (19 - 43)	52.5 (44 - 95)	140 (99 - 340)

- About 25% of patients died within the first month,
- Half of the patients died by around 3 months and
- Only 25% of patients survived beyond approximately 5 months, indicating limited long-term survival.
- The empirical Kaplan–Meier survival curve was closely followed by the generalized gamma curve, suggesting that the model adequately captured the observed survival pattern.

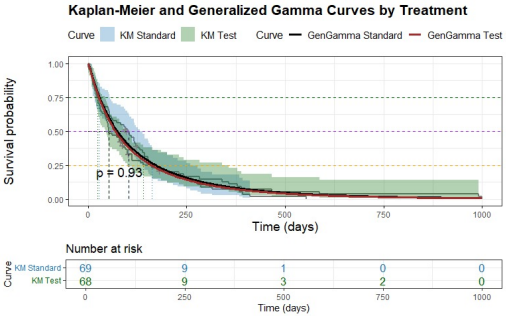

Figure 1: Kaplan–Meier vs Generalized gamma survival curve

Kaplan-Meir survival curve by cell type

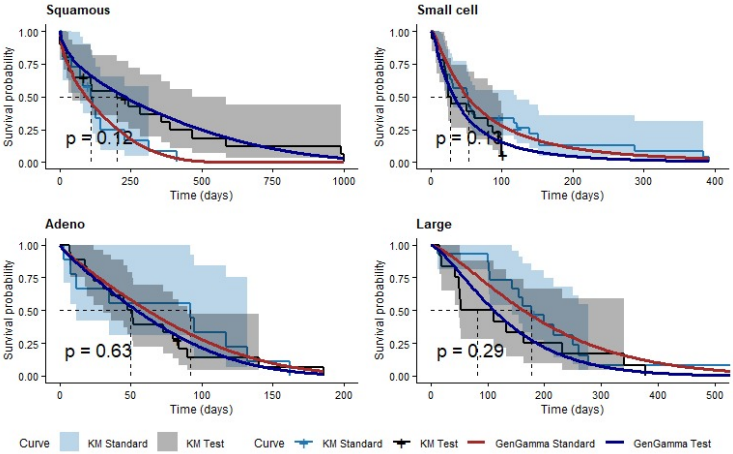

Figure 2: Kaplan-Meir and generalized gamma survival curve by cell type

Non-Parametric Analysis — Nelson-Aalen Estimator

Table 4: Nelson-Aalen Estimator for Standard

time	n.risk	n.event	cumhaz	hazard	CI
3	69	1	0.014	0.014	0.958 - 1
4	68	1	0.029	0.015	0.932 - 1
7	67	1	0.044	0.015	0.91 - 1
8	66	2	0.074	0.030	0.868 - 0.991
10	64	2	0.106	0.031	0.83 - 0.973
11	62	1	0.122	0.016	0.812 - 0.963
12	61	2	0.155	0.033	0.776 - 0.942
13	59	1	0.172	0.017	0.758 - 0.932
16	58	1	0.189	0.017	0.741 - 0.921
18	57	2	0.224	0.035	0.708 - 0.898
20	55	1	0.242	0.018	0.691 - 0.886
21	54	1	0.261	0.019	0.675 - 0.874
22	53	1	0.279	0.019	0.659 - 0.862
27	51	1	0.299	0.020	0.642 - 0.85
30	50	1	0.319	0.020	0.626 - 0.838

Table 5: Nelson-Aalen Estimator for test therapy

time	n.risk	n.event	cumhaz	hazard	CI
1	68	2	0.029	0.029	0.931 - 1
2	66	1	0.045	0.015	0.908 - 1
7	65	2	0.075	0.031	0.866 - 0.991
8	63	2	0.107	0.032	0.828 - 0.972
13	61	1	0.123	0.016	0.809 - 0.962
15	60	2	0.157	0.033	0.773 - 0.941
18	58	1	0.174	0.017	0.755 - 0.93
19	57	2	0.209	0.035	0.721 - 0.908
20	55	1	0.227	0.018	0.704 - 0.896
21	54	1	0.246	0.019	0.687 - 0.884
24	53	2	0.284	0.038	0.654 - 0.86
25	51	3	0.342	0.059	0.605 - 0.823
29	48	1	0.363	0.021	0.59 - 0.81
30	47	1	0.385	0.021	0.574 - 0.797
31	46	1	0.406	0.022	0.558 - 0.784

Nelson-Aalen Curve

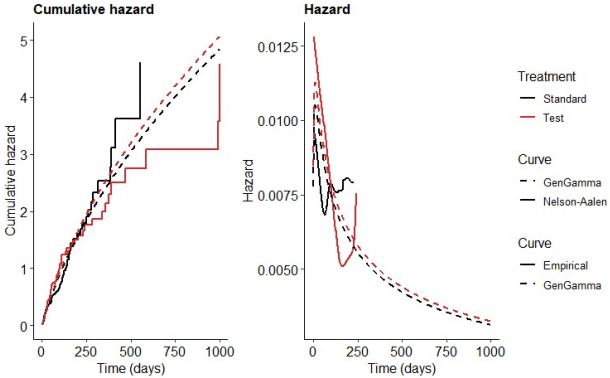

Figure 3: Nelson-Aalen Cumulative hazard and hazard function

The generalized gamma model showed good agreement with the empirical cumulative hazard and captured the overall declining hazard pattern, although minor differences were observed at some time points.

Non-Parametric Hypothesis Testing

Table 6: Log-rank test result

Group	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V	Chi-square	df	p-value
Standard	69	64	64.5	0.0039	0.00823	0.008	1	0.928
Test	68	64	63.5	0.0039	0.00823	0.008	1	0.928

Table 7: Breslow test result

Group	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V	Chi-square	df	p-value
Standard	69	32.2	35.4	0.279	0.871	0.871	1	0.351
Test	68	35.2	32.1	0.308	0.871	0.871	1	0.351

The Log-rank and Breslow tests both showed no significant difference in survival between the Standard and Test groups.

Consistent conclusions

This agrees with the univariable Cox model, in which treatment was also not a significant predictor of survival as seen in figure 4. The difference in p-values arises because Breslow emphasizes early events, while Log-rank gives more equal weight across follow-up time.

Semi-Parametric Analysis — Cox Proportional Hazards Model

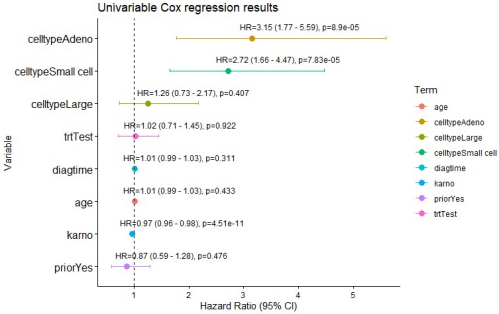

Figure 4: Univariate cox regression result

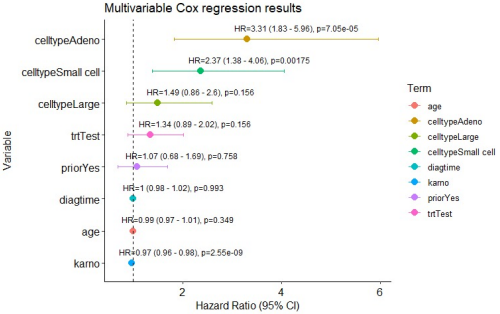

Figure 5: Multivariate cox regression result

Karnofsky score, Adeno cell type, and Small cell type were significantly associated with survival. Higher Karnofsky score reduced the hazard, while Adeno and Small cell types increased the hazard. Other variables were not significant including treatment (figure4). In figure 5 Karnofsky score and cell type remained significant, indicating that their effects are independent of other covariates.

Hazard ratio comparison

Table 8: Hazard ratio comparison

Term	HR (Univariable)	p-value (Univariable)	HR (Multivariable)	p-value (Multivariable)	Absolute Change	%
trtTest	1.018	9.22e-01	1.343	1.56e-01	0.325	31.898
celltypeSmall cell	2.722	7.83e-05	2.367	1.75e-03	-0.355	-13.037
celltypeAdeno	3.151	8.90e-05	3.307	7.05e-05	0.156	4.954
celltypeLarge	1.259	4.07e-01	1.494	1.56e-01	0.235	18.666
karno	0.967	0.00e+00	0.968	0.00e+00	0.001	0.061
age	1.008	4.33e-01	0.991	3.49e-01	-0.016	-1.608
diagtime	1.009	3.11e-01	1.000	9.93e-01	-0.009	-0.898
priorYes	0.867	4.76e-01	1.074	7.58e-01	0.207	23.928

In both univariable and multivariable Cox models, higher Karnofsky score remained protective, while Adeno and Small cell types were associated with worse survival. After adjustment, the hazard ratios changed only slightly; karno (HR: 0.967 to 0.968), Small cell (HR: 2.72 to 2.37), and Adeno (HR: 3.15 to 3.31). Treatment, age, diagnosis time, and prior therapy were not significant.

Proportional Hazards Assumption

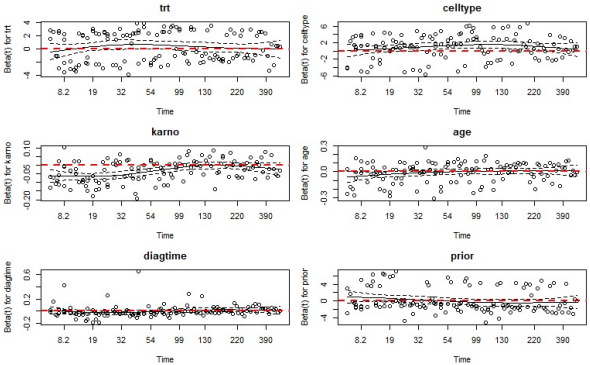

Figure 6: Proportional hazards assumption globally

- The proportional hazards assumption was satisfied for trt ($p = 0.607$), age ($p = 0.176$), diagtime ($p = 0.910$), and prior ($p = 0.141$), but was violated for celltype ($p = 0.0016$) and karno ($p < 0.001$).
- The global test was also significant ($p < 0.001$), indicating that the Cox model does not fully satisfy the proportional hazards assumption overall.

Martingale

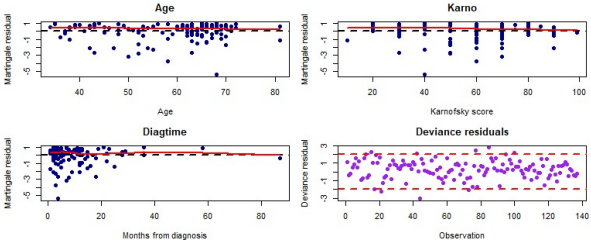

Figure 7: Linearity assumption in log-hazard

- The Martingale residual plots suggest that the continuous covariates have approximately linear effects on the log-hazard.
- Minor departures are visible, especially for age and diagnosis time, there is no strong evidence of nonlinearity.

Model refinement and extended cox models

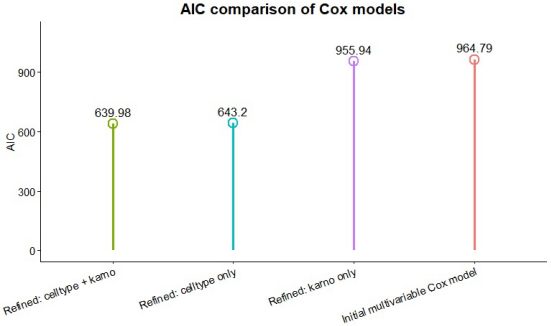

Figure 8: AIC Model refinement comparison

- The initial Cox model included treatment, cell type, Karnofsky score, age, diagnosis time, and prior therapy. In the refined model, cell type was handled through stratification and Karnofsky score was allowed to vary with time.
- Based on AIC, the refined model including both cell type and karno gave the best fit (AIC = 639.98), followed closely by the celltype-only refined model (AIC = 643.20)

Refined Cox Model Results

Table 9: Refined Cox Model Results

Term	Hazard Ratio	95% CI	p-value
Treatment: Test vs Standard	1.232	0.81 - 1.873	3.29e-01
Age	0.988	0.969 - 1.007	2.28e-01
diagtime	0.996	0.978 - 1.014	6.64e-01
Prior therapy: Yes vs No	1.160	0.731 - 1.841	5.29e-01
Karnofsky score	0.924	0.889 - 0.96	5.04e-05
Time-dependent effect of Karnofsky score	1.011	1.001 - 1.022	2.54e-02

In the refined model, each 1-point increase in Karnofsky score lowered the hazard of death initially ($HR = 0.924$), but this protective effect weakened over time.

Discussion

- **Parametric:** The generalized gamma model gave the best parametric fit ($AIC = 1498.94$), showing that survival times were highly variable and the hazard changed over time.
- **Non-parametric:** Kaplan–Meier and Nelson–Aalen showed poor overall survival, with many early deaths and no significant difference between the standard and test groups.
- **Semi-parametric:** Cox regression identified Karnofsky score and cell type as the main predictors of survival, while treatment, age, diagnosis time, and prior therapy were not significant after adjustment.
- **Refined model:** Because PH was violated for cell type and Karnofsky score, the model was refined by stratifying cell type and allowing Karnofsky score to vary with time. This gave the best fit ($AIC = 639.98$).
- The three approaches gave consistent results,survival was poor with many early deaths, treatment groups were similar, and the Cox model identified Karnofsky score and cell type as the main predictors of survival.

Limitation

The study had a relatively small sample size ($n = 137$) and only 9 censored observations, which may limit the precision and generalizability of the results.

Conclusion

- Kaplan–Meier curves declined rapidly early and were closely followed by the generalized gamma model in both treatment groups.
- Nelson–Aalen showed increasing cumulative hazard and an early higher hazard that later declined, consistent with the generalized gamma model.
- There was no treatment effect (Log-rank $p = 0.928$, Wilcoxon $p = 0.351$, Cox $p = 0.922$).
- Survival was poor: median survival was 80 days; about 25% died within 1 month, and only 25% survived beyond about 5 months.
- The hazard was not constant; it was highest early and then declined.
- In the refined Cox model, Karnofsky score remained the key predictor ($HR = 0.924$ initially), while cell type was handled by stratification.
- Clinically, patients with poor performance status and high-risk cell types may need closer monitoring and more individualized care.

Public Implications and Recommendations

Public Implications

- Survival was poor, with many early deaths.
- Treatment groups showed similar survival.
- Karnofsky score and cell type were more important than treatment.
- Further studies are needed to assess whether newer or more targeted treatments could improve survival.

Recommendations

- Assess Karnofsky score early to help identify patients at higher risk.
- Future studies with larger samples are needed to confirm these findings.
- Develop and evaluate better treatment strategies for high-risk patients.
- Use early predictors to support prediction and treatment decisions without waiting for all survival outcomes.

References

[1] Cox, D. R. (1972). *Regression Models and Life-Tables*. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2), 187–220.

[2] Kaplan, E. L., & Meier, P. (1958). *Nonparametric Estimation from Incomplete Observations*. *Journal of the American Statistical Association*, 53(282), 457–481.

[3] Kleinbaum, D. G., & Klein, M. (2012). *Survival Analysis: A Self-Learning Text* (3rd ed.). Springer.

Thank
you!